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India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

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Session : ROLE OF SMALL MODULAR REACTORS (SMR) IN THE NET ZERO POWER SYSTEMS

A Birds Eye View of Indian Scenario & Intrinsic Safety Requirements for SMRs

Presented By

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INTRODUCTION

- Globally SMRs are receiving increased attention (> 80 in 18 countries)
  - All types: Thermal and fast: PWR, PHWR, BWR, LMFR, MSR, HTGR: Floating/underground
  - Two SMRs are already operating (KLT-40S & HTR-PM)
  - SMRs under construction
    - ACP-100 (125 MWe PWR), TMSR-LF1, BREST-OD-300, RITM-200S, RITM-200SN & CAREM
  - SMRs licensed by the regulatory bodies
    - VOYGER 600 (12x50MWe), VOYGER-4 (4x77), VOYGER-6 (6x77MWe) and SMART100 (107 Mwe PWR)
  - SMRs under the licensing process
    - BWRX-300, Rolls-Royce SMR-470, SMR-300, VOYGER-12 (12x77)
  - Some are under detailed design stage
  - Vast majority of SMRS under conceptual design
- Economics - cost/MWe: NC BWR<LMR<HTGR<PWR

Reactor	Cr/MWe
BWRX-300	18.5
LMFR (4S-SFR)	20.7
HTGR (HTR-PM)	44.87
PWR (NuScale 77)	28.8
KLT-40s (2x35)	112.4
CAREM (27)	192.7

SMR Applications

- Proposed SMR applications include
  - Replacement of FFPPs in its own site to reverse global warming (> 100 MWe)
  - Replacement of captive power plants in process industries (10-100 MWe)
  - Solar-Nuclear Hybrid, Remote Communities, Industries & Institutes (up to 10 MWe)
- Other Applications
  - Other applications – Process heat applications – Efficiency 80%
    - District heating, Desalination, Hydrogen generation, Coal gasification/Oil and gas reforming
- Advantages of SMRs include less total cost, mass production in factory
  - Short construction period, smaller site, simplified design, load following
  - Better safety (3000 times safer than commercial NPPs as per IAEA)
  - Transportable, refueling frequency (2 yr – 16 yr)

Indian Scenario

- As part of the proposed 100 GWe by 2047, SMRs and large reactors planned
  - India is a late starter in SMR – At least 5 SMRs by 2033
- Conceptual design of four SMRs by DAE
  - BSR being developed by NPCIL is a 220 MWe PHWR
  - BSMR-200 is being developed by BARC is a 200 MWe PWR
    - Advanced stage of obtaining financial and administrative sanction
  - 55 MWe PWR is another SMR being developed by BARC
    - BSMR-200 and SMR-55 proposed site is Tarapur
  - 5 MWt HTGCR for hydrogen production being developed by BARC
    - Proposed site is Vizag
- Others?

Indian Scenario – Private Sector Participation

- Six Major Private Companies are exploring SMR
  - Adani Power, Reliance Industries, Tata power, Hindalco industries, JSW Energy, Jindal S&P
    - All are collaborating with NPCIL
- In respect of BSR-220, NPCIL floated a RFP (Request for proposals)
  - Last date for the RFP is March 31, 2026
  - 16 Sites identified in six states, SMRs will be built & operated under NPCIL supervision
  - NPCIL retains ownership and operational control & Private firms getting the generated electricity for captive use and will fully finance the projects including capital & lifecycle costs
  - Private firms will reimburse NPCIL for the project execution and decommissioning. In return they will get long-term, reliable power supply for their energy-intensive industrial use.
- There are a few unannounced initiatives
  - 5 MWe liquid metal cooled MMR – IITM initiative

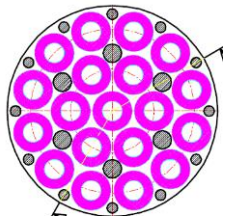
Intrinsic Safety Requirements for SMRs

- SMRs are located in population centres with very small site area
  - SMRs have reduced or no exclusion radius and practically no EPZ
  - SMRs require intrinsic safety features as public domain starts from the site boundary
- Intrinsic safety is usually referred to physics design
  - TRIGA and EBR-II are examples where intrinsic physics safety was demonstrated.
  - Present day reactors with two fast acting shutdown systems safe shutdown is assured
    - TMI-2 and Fukushima caused by failure of thermo-hydraulic systems to remove decay heat
    - Intrinsically Safe Thermal-hydraulic Design (ISTD) is required for SMRs
- Objective of ISTD for SMR? No radioactivity release in public domain
  - Requires assured containment safety: TMI-2 the containment survived.
    - Fukushima Containment was destroyed in hydrogen explosion: Eliminate hydrogen explosion by design
    - Chernobyl did not have a containment: Eliminate operating regions with positive void coefficient

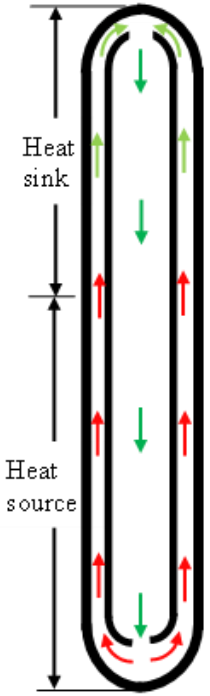
Requirements of ISTD for SMRs?

- **No Loss Of Plant Accident (LOPA): LOPA leads to severe economic loss and bankruptcy to utilities.** TMI-2: no radioactivity release in public domain, but significant core melt led to LOPA.
  - Requirement of ISTD: **No core melt**
- **SMR shall achieve a quasi-steady state operation without operator intervention** with PCT below the threshold value for significant metal-water reaction to occur in water-cooled reactors in case of long-term SBO with LOCA.
  - Requirement: **Two diverse fuel cooling systems (TDFCS) with one being a low pressure IMT-PDHRS**
    - **Patented in Japan, EU and the USA**
  - Other coolants: **Quasi steady state operation with PCT below the core melt temperature**

    in case of loss of heat sink (or LTSBO) : ISTD requirements vary with SMR type
- Several ISTD Concepts have been proposed – TDFCS
- TDFCS: MCS and DPFCS: Requires annular fuel pins and THTD
  - MCS can be active or passive, low pressure or high pressure
  - DPFCS is passive low pressure system



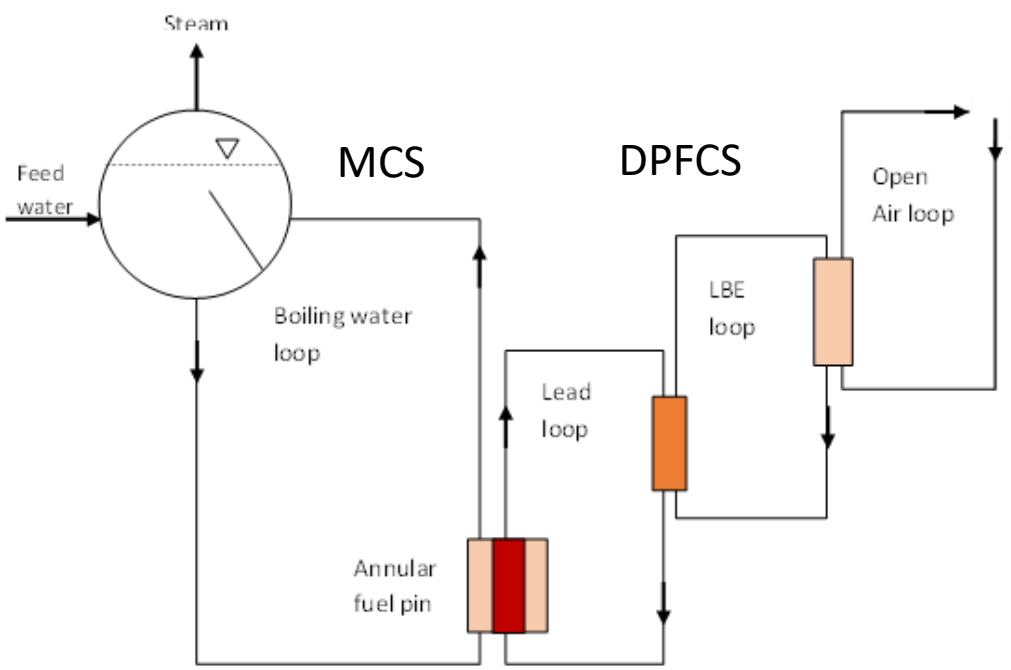
Annular fuel bundle



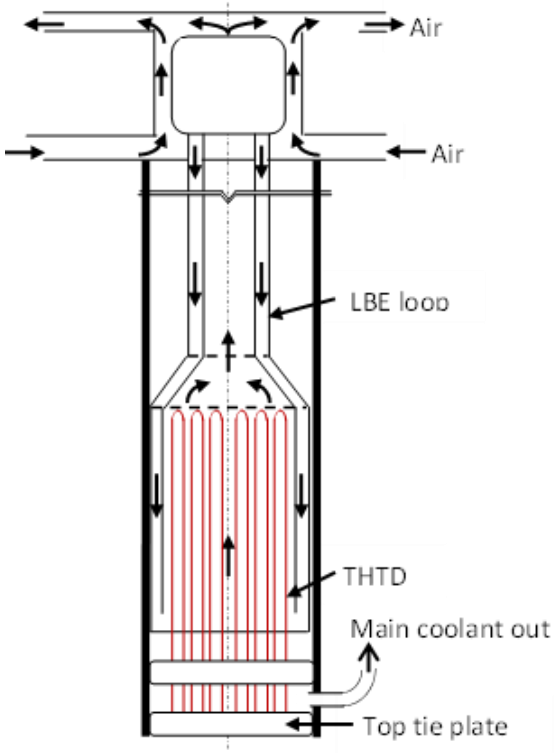
THTD

ISTD Concepts for SMR/ TDFCS for a vertical PTBWR

- Couple NCSs of the Series parallel type
- DPFCS consists of a lead, LBE and open air loops
- DPFCS is essentially an IMT-PDHRS
- MCS: 95% of core heat
- DPFCS: 5%
- DPFCS takes care of decay heat removal even if MCS is dry



Series-Parallel CNCS

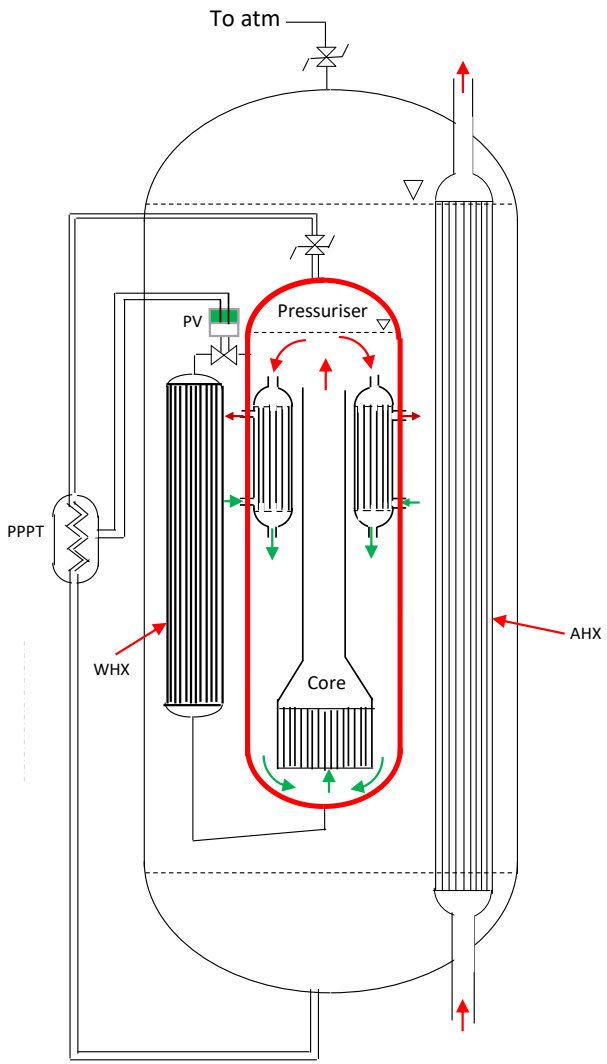


DPFCS

Ref: Vijayan et al. (2018), Proceedings of NURER-2018 and Vijayan et al. (2019), Single-phase, two-phase and supercritical NCSs, Elsevier, 2019.

ISTD Concepts for SMR/ Integral PWR/Concept-1

- Integral PWR with a water-filled secondary containment
- SC has Immersed WHX and AHX bundles
- In case of loss of feed water temperature rises causing pressure rise in RPV
- Opening the relief valve causes the PV to open which opens the WHX circuit.
- At low power the AHX is adequate to remove decay heat.

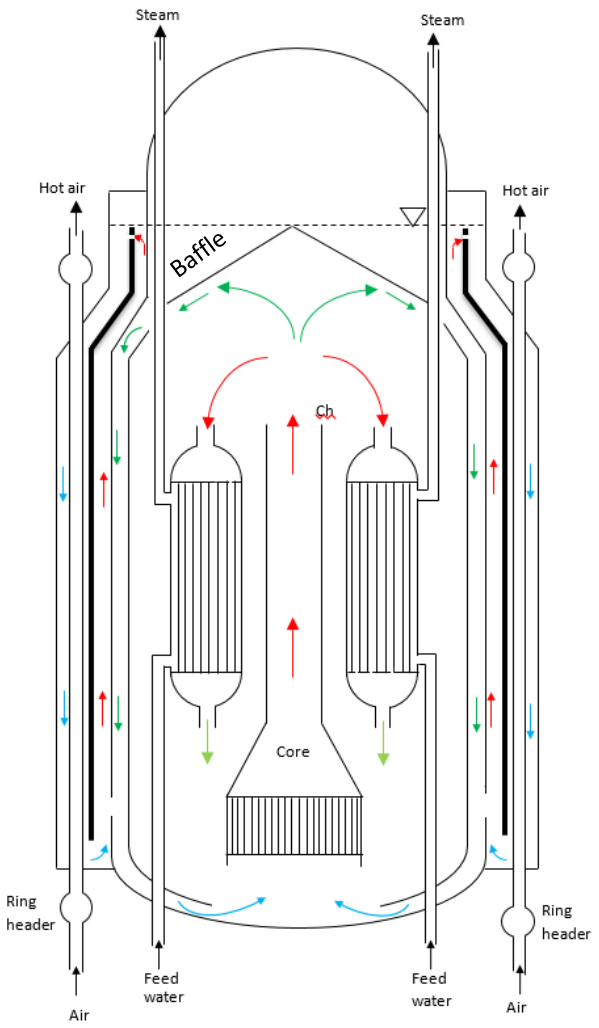


Integral PWR

Ref: Vijayan and Gangwar (2024), Intrinsically safe thermal hydraulic designs for SMRs, IEEE electrification Magazine, 2024

ISTD Concepts for Integral type LM-MMR/ Concept-I

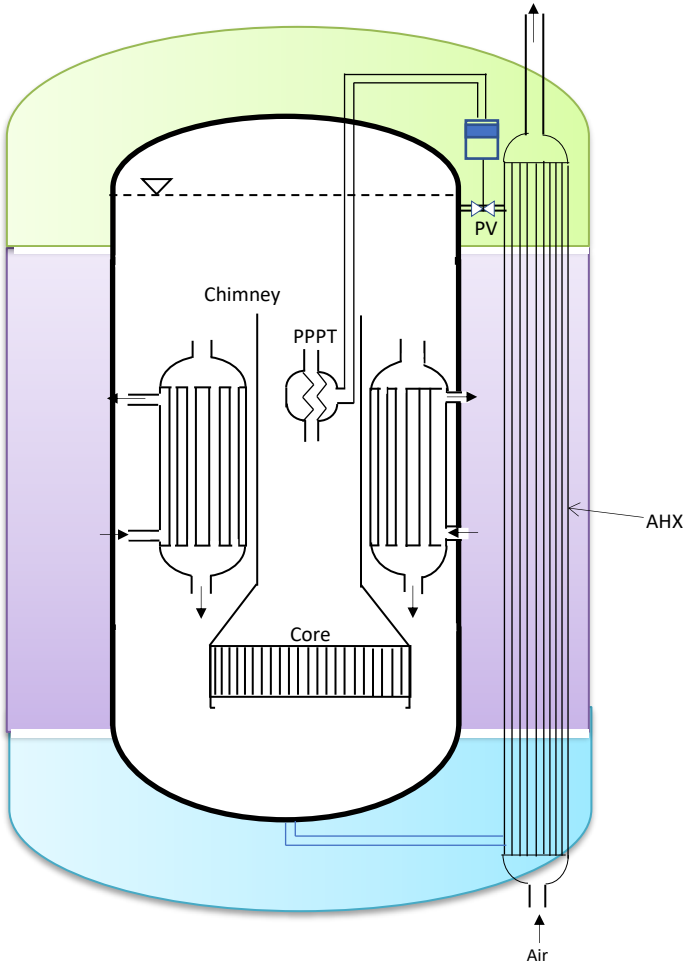
- Reactor vessel with a Jacket having an immersed AHX
- Jacket is open to the main vessel so that the level is same
- Jacket is partitioned to facilitate natural circulation
- Flow restricting orifices are provided at the top of the partition plate.
- Low pressure reactor vessel
- In case of loss of feedwater to the SG, the coolant temperature rises raising the level which increases the flow area and heat removal rate passively.



LM-MMR with Jacket

ISTD Concepts for Integral type LM-MMR/ Concept-II

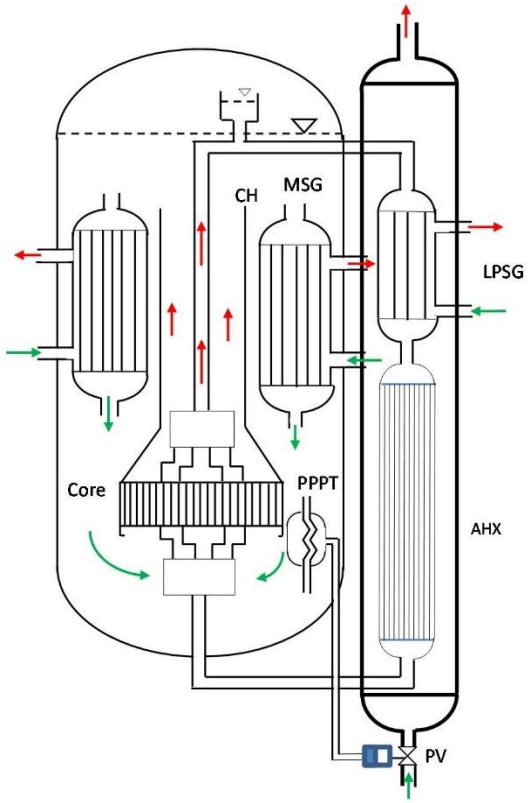
- Secondary containment vessel (SCV) surrounding the RV
- SCV has an immersed AHX open to atmosphere
- In case of loss of feed water to the SG, temperature rises in the chimney initiating PPPT action to open the PV
- As the shell side flow increases, heat removal by AHX increases limiting temperature rise.



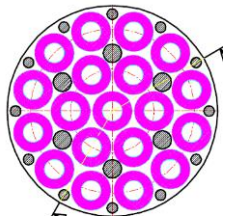
LM-MMR with SCV and AHX

ISTD Concepts for Integral type LM-MMR/ Concept-III

- DPFCS with THTD and annular fuel pin cluster
- MCS and a DPFCS with a hollow cylinder housing the low power steam generator (LPSG) and an AHX tube bundle. The PV at the bottom is closed normally.
- Both MCS and DPFCS generates steam under normal operation with a common turbine
- In case of loss of feed water, the temperature goes up in the RV causing the PPPT to open the PV allowing air into the AHX and the cooling continues.
- DPFCS will function even if MCS is completely dry.



ISTD with DPFCS for LM-MMR



Annular fuel bundle

KEY TAKEAWAYS / RECOMMENDATIONS

- SMR development is spreading all over the world.
- Intense research is ongoing in micro, medium and large power SMRs
- Advantages include enhanced safety, smaller site, capital investment, and construction time
- Applications include the replacement of FFPPs, and captive power plants, hybridization with renewables for RTC operating baseload power plants
- Brief description of Indian SMRs and requirements of ISTD are highlighted
- Different concepts of ISTD are presented which are essentially IMT-PDHRSs eliminating radioactivity release in public domain, hydrogen explosion and LOPA

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Links/References (If any)